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A Design Science Research Approach for Developing an Integrated Information Technology Strategic Planning Framework

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Abstract: Information technology strategic planning is required to align digital investment, enterprise architecture, and governance technology information with the organization's objectives. However, the existing approach is still generally separated into strategic analysis, formulation architecture, and governance control. This research proposes an integrated framework based on Ward & Peppard, TOGAF, and COBIT 2019 using the Design Science Research approach. This research has never been conducted before. The framework is designed to connect analysis, business and systems information / technology information, portfolio applications, enterprise architecture blueprint, governance objectives, capability assessment, risk control, readiness index, and performance measurement. Validation was conducted using expert judgment and the Delphi method with 19 respondents. The calculation resultsshow that the framework is capable strengthen traceability between strategy, architecture, governance, risk, and implementation roadmap, where The strategic alignment dimension has a gap of 7% from **current score of 78 % and the target of 85%**, while risk and control integration has the largest gap. by 19% **from current score of 66 % and of the target of 85%**, dimensions Other aspects, such as Enterprise architecture blueprint, Governance objective mapping, and Capability assessment, have a gap of around 13%, 16% and 15%, while Performance measurement has a gap of 17%. **Overall, the average readiness index across all dimensions reached 71.4%, 79.4% of the target of 85%.** Overall, the average readiness index Expert-Assessed Applicability Index across all dimensions reached 71%of the target of 85%. Contribution to the main study: These are conceptual artifacts that can serve as a basis for preparing the IT Master Plan and the organization's digital transformation.

Keywords: Information technology strategic planning, Ward & Peppard, TOGAF, COBIT 2019, enterprise architecture, IT governance, Design Science Research

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1. Introduction

Information technology strategic planning (IT) has become an important activity for organizations that want to ensure that IT investments support the organization's objectives and generate sustainable income benefits [1]. Digital transformation and the acceleration of new technology adoption will require a structured, capable plan to handle complexity across functions [2]. On the other hand, many organizations still experience misalignment between business strategy and IT initiatives, which results in suboptimal investment outcomes [3]. IT Planning strategy is a crucial element in ensuring harmony between the business organization's objectives and the use of information technology, especially amid

complex digital disruption [4]. In practice, many organizations still face challenges in compiling structured, measurable, and effective IT plans that comprehensively meet business needs. Frequent problems that happen are the existence gap between business strategy and IT strategy, the weakness mapping needs application, not yet optimal IT governance, as well as no integration between strategic planning, enterprise architecture, and control mechanisms. These conditions cause IT initiatives to often run partially, with no good, and not yet a fully value-oriented business. A successful IT planning strategy is heavily influenced by the alignment of business and IT strategies, management

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Commented [Bambang1]: Comment 5 : (Reviewer 1)

The statement that the overall readiness index is "79.4% of the target of 85%" is mathematically incorrect. Dividing the reported mean of 71.4% by the target of 85% gives approximately 84.0%, not 79.4%. If the authors intend to express target attainment, the correct value should be approximately 84.0%. If 79.4% was derived using another formula, that formula and its rationale must be stated explicitly.

Commented [Bambang1R2]: Thank you for this observation. We acknowledge that the previous calculation was incorrectly interpreted. The revised manuscript recalculates the achievement level using the formula:

Target Achievement (%) = (Current Score / Target Score) × 100

Based on this calculation, the achievement level is (71.4/85) × 100 = 84.0%. The incorrect statement has been removed and replaced with the correct interpretation.

Overall, the average readiness index across all dimensions reached 71%of the target of 85%. as related to comment 4

Commented [Bambang2]: Comment 8 : (Reviewer 1)

The abstract states that the enterprise architecture blueprint, governance objective mapping, and capability assessment dimensions have gaps of "around 15%." Their reported gaps are 13%, 16%, and 15%, respectively. Although "around 15%" is approximately defensible, the abstract should preferably report the exact values because these dimensions are later used to support comparative interpretations. Exact reporting would also prevent confusion with the incorrectly reported average gap of 15.6%.

Commented [Bambang2R2]: Thank you for the suggestion. We have revised the Abstract section by replacing approximate expressions with exact quantitative values obtained from the evaluation results.

Abstract: Information technology strategic planning is required to align digital investment, enterprise architecture, and governance technology information with the organization's objectives. However, the existing approach is still generally separated into strategic analysis, formulation architecture, and governance control. This research proposes an integrated framework based on Ward & Peppard, TOGAF, and COBIT 2019 using the Design Science Research approach. This research has never been conducted before. The framework is designed to connect analysis, business and systems information / technology information, portfolio applications, enterprise architecture blueprint, governance objectives, capability assessment, risk control, readiness index, and performance measurement. Validation was conducted using expert judgment and the Delphi method with 19 respondents. The calculation resultsshow that the framework is capable strengthen traceability between strategy, architecture, governance, risk, and implementation roadmap, where The strategic alignment dimension has a gap of 7% from **from current score of 78 % and a the target of 85%**, while risk and control

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support, effective IT governance, and organizational capability [4].

Previous studies show that various frameworks have been used in IT strategic planning. Ward & Peppard are often used to analyze business conditions and systems information, formulate IT strategy needs, and compile portfolio applications. TOGAF ADM is used to build enterprise architecture, including business architecture, data, applications, and technology. While that, COBIT 2019 is used to ensure IT governance and management is running in a way structured through governance objectives, control, measurement of performance, as well as Risk management [5]. In this systematic study of 62 articles from 2015-2024 [4], the Ward & Peppard framework was the most widely used in PSTI, appearing in 29 of the 62 articles (around 47%), followed by TOGAF in 15 of the 62 articles (around 24%). Some frameworks have been widely used to support IT strategy planning. Ward & Peppard present an approach to planning strategic IS/IT that emphasizes analysis of the business environment, formulation of needs, and planning the application portfolio, so that IT planning can be aligned with business objectives [6]. Approach the Lots used in case studies; the latest to implement a combination of SWOT analysis, PEST, CSF, McFarlan grid, and business mapping capability to compile an IS/IT plan [6] [7]. While doing so, TOGAF provides a comprehensive Architecture Development Method (ADM) for designing enterprise architecture (EA) — including business architecture, data, applications, and technology — as well as artifacts (catalogs, matrices, diagrams) that help translate business strategy into a technical blueprint [8]. Empirical studies of more cases show that TOGAF-based EA practices can accelerate organizational transformation when combined with measurement benefits and strong governance [9] [3]. On the governance side, COBIT 2019 sets out comprehensive objectives for governance & management. To ensure value, control risk, and resilience of IT resources [10]. COBIT 2019 is designed to integrate easily with other frameworks (e.g., TOGAF, ITIL, ISO), thereby contextualizing EA for practical governance [10]. Literature review: The latest literature also highlights the role of COBIT in aligning IT strategy with management risk, measuring performance, and governance source power — the essential thing when an organization is implementing an enterprise architecture roadmap [11].

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While each framework has strengths in specific domains, partial implementations of the frameworks still face various limitations. Ward & Peppard is particularly strong in the strategic aspect. Perspective, but it has not explicitly provided an enterprise development mechanism, architecture, and governance. TOGAF ADM excels in enterprise architecture development but places relatively little emphasis on governance mechanisms, monitoring, and comprehensive IT performance evaluation. On the other hand, COBIT 2019 has a strong orientation toward governance and management information, as well as toward technology companies. Although COBIT 2019 includes APO02 *Managed Strategy* and APO03 *Managed Enterprise Architecture*, this framework does not provide a detailed and comprehensive methodology for formulating business strategies from analysis of the environment until determining portfolio priorities, or for developing *baseline architecture*, *target architecture*, analyzing gaps, and mapping the road migration as provided by the strategic planning methodology and TOGAF ADM [10] [12] [8].

The state of the art shows increasing attention to integrating strategic planning, enterprise architecture, and IT governance to support an organization's digital transformation. Several studies have attempted to combine enterprise architecture with a governance framework to improve the effectiveness of IT implementation and organizational change management [13][14][15][16]. The literature shows that enterprise architecture and IT governance play complementary roles, but integrating them into a single validated operation remains limited [17][18][19][20].

Based on these conditions, a research gap remains: the lack of an integrated conceptual framework that simultaneously incorporates Ward & Peppard, TOGAF ADM, and COBIT 2019 into a single adaptive strategic IT planning model. Research has previously discussed the planning framework, strategy, enterprise architecture, and IT governance separately or in partial combination. The study's IT planning strategy identified Ward and Peppard, TOGAF, and COBIT as relevant approaches but has not yet demonstrated a mechanism to orchestrate all three into a single integrated model [4]. These limitations have the potential to hinder the maintenance of business-IT alignment, digital transformation coherence, and systematic, evidence-based IT performance measurement [21][22][23].

To address this gap, this study proposes a conceptual framework called PRASASTI (Integrated

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Strategic IT Planning Framework), a hybrid model that integrates Ward & Peppard, TOGAF ADM, and COBIT 2019 into a single, unified IT strategic planning framework. In the proposed framework, Ward & Peppard are used as the strategic perspective to analyze business needs and organizational priorities, TOGAF ADM is used as an architectural perspective for building an enterprise architecture systematically, while COBIT 2019 is used as a governance perspective to ensure monitoring, evaluation, control, and realization of IT benefits run continuously.

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The main contribution of this research is the development of a multidimensional conceptual model that links the domains of strategy, enterprise architecture, and IT governance into a single, integrated cycle. strategic IT planning. This research also produces a mapping of the relationships among frameworks to clarify each domain's contribution to supporting strategic IT planning, organizational alignment. Furthermore, this strategic planning framework serves as an adaptive conceptual foundation for developing and implementing IT master plans in modern organizations. This study uses a design science research approach because its main objective is to develop conceptual artifacts in the form of a framework. [24] Explain that design science research is relevant for studying system information because it is oriented towards creating and evaluating artifacts that can give marked practical and scientific value.

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The remainder of this paper is organized as follows. Section 2 presents the theoretical background related to IT strategic planning, enterprise architecture, and IT governance frameworks. Section 3 discusses the research gap and novelty of the proposed PRASASTI framework. Section 4 describes the Design Science Research methodology used for artifact development and validation. Section 5 presents the proposed framework and its integration mechanism. Section 6 discusses the validation results, expert assessment, and gap analysis. Finally, Section 7 presents the discussion, research limitations, and conclusions.

2. Literature Review

2.1. Planning IT Strategy

IT strategy planning is a systematic process for identifying business needs, evaluating current IS/IT conditions, determining priority applications, designing infrastructure, and preparing an implementation roadmap. In practice, planning an IT

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strategy must answer questions about the system the organization needs, how the system supports business processes, what technology is required, and how IT investments deliver value to the organization.

[25] Explain that planning strategic system information is required when system information utilization is not yet at maximum capacity, as the document planning strategy does not yet exist and will serve as a reference for development. This strengthens the argument that planning an IT strategy is not merely a list of projects, but must be based on an analysis of the organization's needs and priorities.

2.2. Ward & Peppard

Ward & Peppard emphasize that the IS/IT strategy must start with a business approach that covers analysis of the internal and external business environments, as well as the internal and external IS/IT environments. The main outputs include system strategy information, technology strategy information, IS/IT management strategy, and future portfolio applications.

Study [26] applies Ward & Peppard to e-commerce companies and shows that this approach can produce a recommendation portfolio application. The study confirms that SWOT analysis, Value Chain, PEST, CSF, and McFarlan Strategic Grid are important tools for strategic planning in IS/IT. According to [27], Ward & Peppard can help organizations that do not have their own integrated systems formulate the need for IS/IT development applications and strategies.

Ward & Peppard's limitations lie in their not yet having provided detailed mechanisms for translating the IS/IT strategy into a complete enterprise architecture. Therefore, this framework needs to be equipped with enterprise architecture approaches such as TOGAF.

Commented [Bambang3]: Respon to comment 1 from reviewer 1

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2.3. TOGAF and Enterprise Architecture

Enterprise architecture is an approach to describing the connections among business processes, data, applications, technology, and infrastructure within an organization. TOGAF provides an Architecture Development Method for compiling the architecture vision, business architecture, data architecture, application architecture, technology architecture, opportunities and solutions, migration planning, implementation governance, and architecture change management.

According to [28] enterprise architecture can enable digital transformation by helping organizations align IT strategy, technology, processes, and business objectives. Meanwhile, according to [29], TOGAF 10 can be used to support digital transformation in organizations, especially small and medium-sized ones, through more enterprise architecture design.

Although TOGAF is strong in architectural blueprint preparation, it does not yet approach this in a way that specifically provides mechanism analysis for portfolio application in business strategy, as proposed by Ward & Peppard. In addition, TOGAF does not provide capability assessment and governance objectives in a comprehensive, explicit manner, as COBIT 2019

2.4. COBIT 2019 and IT Governance

COBIT 2019 is an IT governance and management framework that is oriented towards creating value, managing risk, optimizing resource power, and measuring performance. COBIT 2019 domains include: Evaluate, Direct, and Monitor; Align, Plan, and Organize; Build, Acquire, and Implement; Deliver, Service, and Support; and Monitor, Evaluate, and Assess.

[30] Using COBIT 2019 to assess the capability level of the system information library. Research

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findings indicate that COBIT 2019 can help assess the current state, set targets, and develop recommendations to improve IT governance. [31] confirms that the COBIT 2019 Design Factor Toolkit can help map governance objectives and produce gap scores as a basis for IT governance improvements.

COBIT 2019's strengths lie in its governance objectives, capability assessment, risk control, and performance measurement. However, COBIT 2019 is not a primary framework. For a comprehensive analysis of strategic IS/IT and enterprise architecture blueprint. Therefore, COBIT 2019 is necessarily integrated with Ward & Peppard and TOGAF.

2.5. Framework Integration

A number of recent studies have combined the two frameworks, but few have yet integrated Ward & Peppard, TOGAF, and COBIT 2019 into a single end-to-end model. [32] Adaptation planning, strategic, and information systems with COBIT 2019 at universities, post-COVID-19. Research shows that COBIT 2019 can be used to determine governance domains and to conduct capability gap analysis after the strategic planning process is complete.

[33] Integrating COBIT 2019 and TOGAF ADM for designing enterprise architecture based on risk technology information. The integration is relevant because it can connect IT risks to enterprise architecture design. However, the approach is not yet explicitly included in Ward & Peppard as a baseline analysis for strategic IS/IT and portfolio applications.

Therefore, this research fills the gap by integrating three frameworks: Ward & Peppard as the basis for strategic IS/IT analysis, TOGAF as the basis for preparing the enterprise architecture blueprint, and COBIT 2019 as the governance principles, capabilities, risks, controls, and performance measurement.

Table 1. Summary study previously

Reference	Summary findings and gaps
Andry et al. (2023)	Ward & Peppard produces portfolio application for e-commerce; not yet covers enterprise architecture and governance.
Desyana & Sitokdana (2022)	Ward & Peppard helps identification IS/IT based needs analysis business; not yet measure governance capabilities.
Harits et al. (2022)	Ward & Peppard and COBIT 2019 support capability gap analysis; not yet generate TOGAF blueprints.
Rahimi et al. (2023)	Enterprise architecture becomes an enabler of digital transformation; not yet focus on governance objectives.
Safitri et al. (2024)	TOGAF 10 supports SME digital transformation; not yet integrate COBIT 2019 full.
Christian et al. (2023)	COBIT 2019 assesses capability levels and recommendations repair; not yet connect portfolio application.

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Reference	Summary findings and gaps
Mubarak & Fianty (2023)	The COBIT 2019 Design Factor Toolkit determines governance priorities ; not yet building enterprise architecture.
Putra et al. (2025)	TOGAF and COBIT 2019 integrate IT architecture and risks ; not yet enter Ward & Peppard.

To further justify the identified research gaps, a comparative analysis was conducted among existing IT strategic planning frameworks, including Ward and Peppard, TOGAF, COBIT 2019, and previous framework integration approaches. The comparison was performed based on several critical dimensions required in comprehensive IT strategic planning, including strategic analysis, application portfolio management, enterprise architecture development, IT governance, capability assessment, risk management, and performance measurement. This analysis aims to identify the strengths and limitations of each framework and demonstrate the necessity of developing an integrated framework that addresses the gaps between strategic planning, architecture design, governance, and continuous performance evaluation.

Table 2 . Comparative Analysis of Existing IT Strategic Planning Frameworks

Capability Dimension	WP	TOG	COB	PRA
Business strategy analysis	S	P	P	I
IT requirement identification	S	P	L	I
Enterprise architecture blueprint	L	S	L	I
Application portfolio prioritization	S	P	P	I
Governance objectives	L	P	S	I
Capability assessment	L	P	S	I
Risk and control	L	P	S	I
Performance measurement	L	P	S	I
Requirement-to-control traceability	X	P	P	PM
Priority propagation mechanism	X	X	X	PM
Cross-framework feedback loop	X	P	P	PM

Noted :
WP=Ward & Peppard;
TOG=TOGAF;COB=COBIT 2019; PRA=Prasasti

S = Strong ; P=Partial; L=Limited; I=Integrated;
PM=Proposed Mechanisme ; X=not exist

The table no. 2, presents a comparative analysis of capability dimensions among Ward and Peppard, TOGAF ADM, COBIT 2019, and PRASASTI. The comparison shows that each existing framework provides strong capabilities in specific areas: Ward and Peppard focuses on strategic analysis and IT portfolio planning, TOGAF ADM emphasizes enterprise architecture development, while COBIT 2019 provides governance, control, capability assessment, and performance management. However, the comparison highlights several integration gaps in existing frameworks, particularly in requirement-to-control traceability, priority propagation, and cross-framework feedback mechanisms. PRASASTI addresses these limitations by introducing an integrated mechanism that connects strategic requirements, enterprise architecture decisions, governance objectives, and continuous improvement processes. Therefore, PRASASTI extends beyond framework combination by providing structured integration and traceability across IT strategic planning domains.

3. Research Gap and Novelty

Based on the library review, there are three major research gaps. First , the integration of Ward & Peppard and TOGAF is strong. In IS/IT strategy analysis and enterprise architecture, it does not yet explicitly provide governance objectives, capability assessment, risk control, and performance measurement. Second, the integration of Ward & Peppard and COBIT 2019 is strong in IS/IT strategy and governance, but has not yet produced a complete enterprise architecture blueprint. Third, the integration of TOGAF and COBIT 2019 is strong in architecture and governance, but not yet strong enough in strategic analysis of IS/IT and in compiling an application portfolio aligned with business strategy.

Research novelty: This lies in the development of an integrated framework that combines Ward & Peppard, TOGAF, and COBIT 2019 into a single planning model for IT strategy. This framework connects business strategy analysis, portfolio application, enterprise architecture blueprint, governance objectives, capability assessment, risk control, readiness index, gap analysis, and performance measurement. Additionally, the

Commented [Bambang4]: Comment 2 (From Reviewer 1)

2. The authors' proposed method does not adequately describe their data. The drawback and limitation described in section 2 are not supported by any demonstrations. Readers will fail to understand the scientific contribution of this research. The authors should justify it by demonstrating specific data in the comparison part.

Commented [Bambang4R2]: Comment 1 (From Reviewer 2) :

The manuscript does not contain a head-to-head quantitative comparison against any baseline. Ward and Peppard, TOGAF, COBIT 2019, and prior two-framework integrations are discussed descriptively, but they are not applied to the same organization, requirements, evaluators, or outcome criteria. Consequently, the study cannot determine whether PRASASTI improves strategic alignment, artifact completeness, governance coverage, planning time, decision quality, or implementation success. The authors should evaluate PRASASTI and at least three credible baselines under an identical case protocol. Relevant baselines should include TOGAF alone, Ward and Peppard combined with TOGAF, and TOGAF combined with COBIT 2019. The comparison should report predefined quantitative measures, uncertainty estimates, and qualitative implementation costs.

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research uses the Design Science Research approach [24] to build conceptual artifacts and the Delphi method [34][35] to validate the model through expert judgment.

Moreover, this integrated framework provides a comprehensive approach to IT strategic planning by unifying strategy analysis, enterprise architecture, and governance objectives within a single model. It enhances alignment between business goals and IT initiatives while facilitating effective resource allocation and portfolio prioritization. The framework also incorporates readiness and gap analysis, allowing organizations to identify weaknesses and implement corrective measures proactively. Through performance measurement and capability assessment, the model supports continuous improvement and informed decision-making. The inclusion of expert validation through the Delphi method enhances the framework's reliability and applicability. Overall, this research contributes a practical and theoretically grounded approach for organizations seeking structured, integrated IT strategic planning.

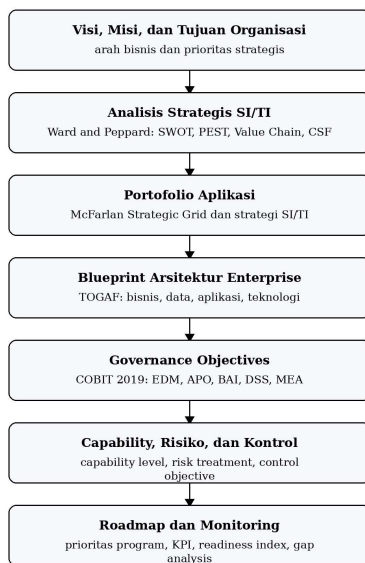


Figure 1 Conceptual model of the PRASASTI Framework

In the first stage, the organization's vision, mission, and goals form the basis for strategic analysis. Second stage using Ward & Peppard to evaluate the business environment and IS/IT. The third stage produces a portfolio application strategy.

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Fourth stage: using TOGAF to build architecture for business, data, applications, and technology. Fifth stage mapping COBIT 2019 governance objectives. Sixth stage measuring capability levels, risks and controls. Seventh stage: prepare the implementation roadmap, KPI, readiness index, and gap analysis.

4. Proposed Framework

Proposed framework named PRASASTI Framework (Integrated Strategic, Architecture and Governance Framework for IT Strategic Planning). This model consists of seven main stages: organizational strategy analysis, analysis of strategic IS/IT, formulation of portfolio applications, enterprise architecture blueprint preparation, governance objectives mapping, capability assessment and risk control, as well as roadmap and performance measurement.

Table 2. Components of the PRASASTI Framework

Component	Sources and functions
Strategic analysis	Ward & Peppard: Identifying need business and IS/IT direction
Application portfolio	Ward & Peppard: Defining priority future applications
Business architecture	TOGAF: Mapping business processes and capabilities organization
Data architecture	TOGAF: Designing data structure, flow, and ownership
Application architecture	TOGAF: Defining application targets and integration system
Technology architecture	TOGAF: Designing platforms, networks, infrastructure, and security
Governance objectives	COBIT 2019: Connecting objective organization with IT governance objectives
Capability assessment	COBIT 2019: Measuring current and target capability levels
Risk and control	COBIT 2019: Managing IT risks and controls
Performance measurement	COBIT 2019: Defining KPIs and monitoring roadmap

Table 2 above shows that the PRASASTI framework was built by integrating three main approaches: Ward & Peppard, TOGAF, and COBIT 2019. Ward & Peppard is used in the initial component, as a *strategic analysis* and *application portfolio*, to identify business needs and IS/IT direction and to determine future application priorities. Thus, the initial phase of this framework

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serves as a basis for aligning organizational strategy with the needs of information systems development.

Furthermore, TOGAF is used to strengthen enterprise architecture by integrating business components, architecture, data architecture, application architecture, and technology. These four components help organizations map business processes, data structures, application integration, and technology infrastructure needs. Meanwhile, COBIT 2019 complements the framework by addressing IT governance, capability measurement, risk management, control, and KPI determination and monitoring. The integration of these three frameworks makes PRASASTI a more comprehensive IT strategic planning framework, as it covers strategy, architecture, governance, risk, and performance measurement.

Table 2. Variables, Indicator and references

Variables	Indicators and references
Strategic alignment	Compliance objective business and IT (Ward & Peppard)
Strategic analysis	SWOT, PEST, Value Chain, CSF (Ward & Peppard)
Application portfolio	McFarlan Strategic Grid and priorities application (Ward & Peppard)
Enterprise architecture	Architecture business , data, applications , technology (TOGAF)
Governance objectives	Mapping objective organization to governance objectives (COBIT 2019)
Capability assessment	Current capability and target capability (COBIT 2019)
Risk control	Identification risk , control , and mitigation (COBIT 2019)
Performance measurement	KPI, readiness index, and monitoring (COBIT 2019)

5. Research Methodology

This study uses a design science research approach . This approach was chosen because the study aims to produce conceptual artifacts in the form of a planning framework for IT strategy that can be used to solve practical problems in organizations. Stages of the study covering identification of the problem, determination of the solution, design and development of objectives, artifacts, demonstrations, evaluation, and communication, as shown in the figure below.

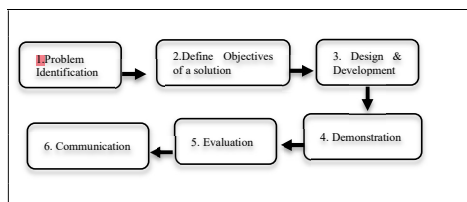


Figure 2. Stages Methodology Research

Data collection techniques include literature studies, framework analysis, instrument compilation, expert judgment questionnaire, Delphi round 1, Delphi round 2, and documentation comment expert. Instrument use Likert scale (1 to 5) [36] to evaluate the relevance, clarity, completeness, measurability, and feasibility of implementation for every dimension and indicator of the framework.

This study also uses expert judgment with the Delphi method to evaluate and validate the developed results, content validity ratio, and Aiken's V [37][38] . The reliability of an instrument can be assessed using Cronbach's alpha.

5.1. formula Testing

Validity content can be counted using the content validity ratio and Aiken's V. Reliability of the instrument can be counted using Cronbach's alpha. Delphi consensus can be analyzed using the interquartile range. Index readiness is used to implement the count percentage readiness for every framework dimension. Validity content can be calculated using the content validity ratio and Aiken's V. Reliability of the instrument can be calculated using Cronbach's alpha. Delphi consensus can be analyzed using the interquartile range. In this study, the expert-assessed applicability index is used to measure the perceived completeness, suitability, and implementation feasibility of each PRASASTI framework dimension based on expert evaluation. This index does not represent organizational maturity or actual organizational readiness.

As for the formulation for content validity ratio:

$$CVR_i = (n_e - N/2) / (N/2) \quad (1)$$

Information :

Respondents' validation was designed for up to 19 people, including expert planning IT strategists, enterprise architecture experts, IT governance

Commented [Bambang6]: This is for the update sentences for comment 2 , from reviewer 2

Commented [Bambang5]: Comment 7 : Reviewer 2 :

The expert-validation design may contain dependence and confirmation biases that are not adequately controlled. The manuscript does not report whether experts were anonymous to one another, whether the framework developers participated in validation, how conflicts of interest were managed, or whether experts independently evaluated competing frameworks. Because the evaluation instrument appears to have been designed around the proposed framework's own dimensions, high agreement may partly reflect criterion alignment rather than comparative merit. The study should include independent assessors, blinded artifact evaluation where feasible, balanced baseline descriptions, preregistered criteria, and analysis of agreement by expert category.

Commented [Bambang5R2]: Thank you for raising this important concern.

The Delphi validation process was conducted through independent expert assessments to minimize individual influence and confirmation bias. Experts were selected based on their experience in IT strategic planning, enterprise architecture, IT governance, and information systems research. The iterative Delphi process allowed refinement of assessment items based on collective expert feedback.

5

experts, IT practitioners, and academic system information experts. Amount: This is used for expert judgment and Delphi,

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n_s = number the expert who rated the item as important
 N = number of all experts

3. Variables and indicators study

36

The formulation of Aiken's V is:

$$V = \Sigma(s) / [n(c - 1)] \quad (2)$$

Information :

$$s = r - l_o$$

r = score given by expert

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l_o = lowest score

n = number of experts

c = number of assessment categories

The Cronbach's Alpha formula is :

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$$\alpha = \frac{k}{(k - 1)} \left(1 - \frac{\Sigma \sigma_i^2}{\sigma_t^2} \right) \quad (3)$$

Description :

k = number of items

σ_i^2 = item variance

σ_t^2 = total variance

As for the formulation Reading Index:

$$\text{Index} = (\text{Actual Score} / \text{Maximum Score}) \times 100\% \quad (4)$$

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All statistical indicators, including Content Validity Ratio (CVR), Content Validity Index (CVI), Aiken's V, Cronbach's Alpha, Kendall's W, and IQR, were calculated using the original numerical values. For presentation purposes, the results were rounded consistently to two decimal places to improve readability while maintaining calculation accuracy.

6. Results

As for the results discussion in study This is as following :

6.1 Problem Identification

The first stage is problem identification. At this stage, the researcher analyzes the main problem that occurred in the object of research. In the study of the development of a planning framework, the main IT strategic issue identified is that many organizations do not yet have an integrated IT planning framework. IT strategy planning is often still done in a partial

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way, for example, by only compiling a list of applications, only creating architectural blueprints, or only conducting an IT governance audit without connecting it to the business strategy and organization.

At this stage, the researcher finds that Ward and Peppard's approach is strong in analyzing strategic IS/IT and portfolio applications, but does not yet provide a detailed design of enterprise architecture in general. TOGAF is strong in compilation architecture, business , data, applications, and technology, but not yet in ways that provide specialized strategic analysis and application portfolio approaches, such as those of Ward and Peppard. Meanwhile, COBIT 2019 is strong in IT governance, capability assessment, risk control, and performance measurement, but not the main framework for developing IS/IT strategies and enterprise architecture blueprints. Therefore, the main problem is not yet the lack of a framework that integrates aspects of strategy, architecture, governance, risk, and performance measurement.

6.2 Define objectives of a solution

The stage two is to define the objective of the solution to be developed. The goal of the solution is to develop a new framework that integrates three main frameworks: Ward and Peppard, TOGAF, and COBIT 2019.

Solution goals that can be explained in terms of a number of targets. First, the framework must be capable of helping organizations conduct strategic business analysis and IS/IT analysis. Second , the framework must be capable of producing a portfolio-appropriate application for the organization's needs. Third , the framework must be capable of translating IS/IT strategy into an enterprise architecture blueprint. Fourth , the framework must provide a mechanism for mapping governance objectives and capability assessment. Fifth, the framework must include aspects of risk control, performance measurement, a readiness index, and gap analysis.

Commented [Bambang7]: This is updated , related comment 11 from reviewer 1

6.3 Design and development

The third stage is design and development, namely the process of developing artifacts and research . This framework is the result of integrating three main sources: Ward and Peppard for strategic analysis and application portfolio aspects; TOGAF for enterprise architecture aspects; and COBIT 2019 for aspects of governance, capability, risk, control, and performance measurement. At this stage, the researcher begins compiling the framework structure. Ward and Peppard's components are used

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in the initial framework for identifying the need for business, IS/IT direction, internal and external conditions of the organization, as well as priority future applications. TOGAF components are used to compile business, data, application, and technology architecture. COBIT 2019 components are used to map IT governance objectives, measure process capability, manage risk, determine controls, and compile KPIs.

The stages of design and development of the framework artifact can be depicted in Figure 3 below

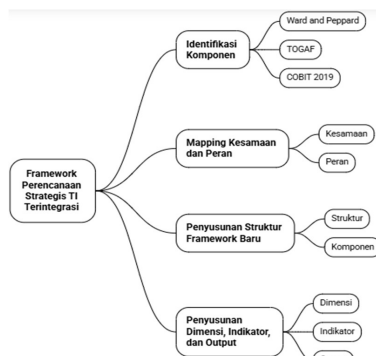


Figure 3. Artifact Framework Flowchart

As for identification component from each framework can depicted in form table under This :

Table 4. Mapping Framework Components in the Design and Development Stage

No	Source Framework	Main Components	Function in the New Framework	Output Generated
1	Ward and Peppard	Strategic analysis	Identifying need business , direction strategic organization , opportunities , threats , strengths and weaknesses related to IS/IT	Analysis need business and IS/IT strategy direction
2	Ward and Peppard	Internal business analysis	Analyze business processes , resources power , structure organization , and needs internal information	Business process and requirements map internal information
3	Ward and Peppard	External business analysis	Analyze factor external like regulation , competition , technology , and stakeholder needs	Analysis environment external organization
4	Ward and Peppard	Internal IS/IT analysis	Evaluate condition applications , infrastructure , data, IT human resources, and IT services that are currently being walk	Condition existing IS/IT
5	Ward and Peppard	External IS/IT analysis	Identifying trend technology and practice best relevant with organization	Opportunity adoption technology new
6	Ward and Peppard	Application portfolio	Determine priority future applications based on mark strategic and needs organization	Portfolio future applications
7	TOGAF	Architectural vision	Developing a vision harmonious architecture with objective business and IT strategy	Enterprise architecture vision
8	TOGAF	Business architecture	Mapping business processes , functions organizations , services , actors , and capabilities business	Architectural blueprint business
9	TOGAF	Data architecture	Designing data structures , data requirements , data flows , and integration information	Data architecture blueprint
10	TOGAF	Application architecture	Designing connection between applications , integration systems and support application to business processes	Architectural blueprint application
11	TOGAF	Technology architecture	Designing need infrastructure , network , platform, security , and technology supporters	Architectural blueprint technology
12	TOGAF	Migration planning	Determine stages migration from condition moment This going to target condition	Migration roadmap architecture
13	COBIT 2019	Governance objectives	Map objective organization to IT governance objectives	Governance objectives map
14	COBIT 2019	Capability assessment	Measure current IT process capabilities this and determine capability targets	Current capability and target capability values

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15	COBIT 2019	Risk and control	Identifying IT risks , controls , and mitigations required	IT risk , control and mitigation register
16	COBIT 2019	Performance measurement	Determine KPIs, monitoring indicators , and measures success roadmap implementation	KPIs and indicators IT performance

As for mapping and similarities and roles from each framework can depicted in form table under This :

Table 5. Mapping similarities and roles from each framework at the design and development stage

No	Integration Aspects	Ward and Peppard	TOGAF	COBIT 2019	Roles in the New Framework
1	Orientation main	Planning strategic IS/IT	Enterprise Architecture	IT governance and management	Combining IT strategy, architecture , and governance in one integrated framework
2	Alignment business and IT	Aligning business strategy with IS/IT strategy	Aligning architecture business , data, applications , and technology	Aligning objective organization with governance objectives	Ensure all over IT initiatives support objective organization
3	Analysis condition organization	Analyze environment internal- external business and IS/IT	Identifying baseline architecture and target architecture	Using enterprise goals and design factors	Become base for understand condition moment this and the need change
4	Planning application	Building a portfolio application moment this and the future	Designing application architecture	Arrange control and management application	Determine priority aligned applications with architecture and governance
5	Design architecture	Not designing architecture in a way details	Developing business, data, application, technology architecture	Supporting governance of IT architecture and services	TOGAF becomes basic enterprise architecture blueprint
6	IT governance	Developing IS/IT management strategies	Supporting architecture governance	Providing governance and management objectives	COBIT 2019 becomes base IT control , measurement , and governance
7	Measurement capability	Does not provide a detailed process capability model details	Support evaluation readiness architecture	Provide capability assessment	Measure condition moment this , targets, and IT capability gaps
8	Risk and control	Identifying risk strategic through analysis environment	Consider risk in design architecture	Provide risk management and control objectives	Ensuring IT strategy and architecture have control and mitigation risk
9	Implementation roadmap	Produce IS/IT strategic direction and priorities application	Providing migration planning	Provide KPIs and monitoring mechanisms	Produce a targeted and measurable IT implementation roadmap
10	Main output	IS/IT strategy and portfolio application	Enterprise architecture blueprint	Governance objectives, capability level, risk control, and KPI	Produce IT Master Plan, digital transformation roadmap , EA blueprint, and IT governance model

Result of stage This can in the form of table integration framework, framework flowchart , conceptual model , indicators measurement , instruments validation , as well as formula calculations used . As seen in table under this related

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with mapping integration based on order framework development .

Table 6. Integration Mapping of Ward and Peppard, TOGAF, and COBIT 2019 into the Proposed PRASASTI Framework

No	IT Strategic Planning Stage	Ward & Peppard Contribution	TOGAF ADM Contribution	COBIT 2019 Contribution	Integration Mechanism / Transformation	Output Artifact	Dimension
1	Initiation and Preparation	Identify initial business needs and strategic objectives	Preliminary Phase: define architecture principles, scope, and stakeholders	EDM01, APO01: governance framework and management objectives	Establish governance structure, scope, principles, and planning mechanism for IT Masterplan development	Project Charter, Scope, Governance Structure	Strategic Alignment
2	Business Strategy and Requirement Analysis	Business analysis, internal/external analysis, CSF, SWOT, Value Chain	Phase A Architecture Vision: stakeholder needs and enterprise vision	EDM02, EDM05, APO02: ensure business and IT alignment	Transform business objectives into strategic IT requirements	Business Requirement Map, IT Strategic Requirement	Strategic Alignment
3	Current SI/IT Assessment	Internal analysis, application portfolio, technology assessment	Baseline Architecture (Current State Architecture)	APO03, APO14, BAI09, BAI10, DSS01: assessment of assets, data, and services	Integrate current application, data, technology, and capability conditions	Current State Architecture (As-Is)	IS/IT Portfolio Planning
4	Target Identification	Future direction and strategic options	Target Architecture Definition	COBIT Design Factors, Enterprise Goals, Alignment Goals	Determine target conditions based on standards, best practices, and organizational needs	Target Capability and Target Architecture	Enterprise Architecture Blueprint
5	Gap Analysis	Strategic gap analysis between current condition and business requirements	Architecture Gap Analysis	APO11, APO12, MEA01: capability, risk, and performance gaps	Compare As-Is and To-Be conditions to determine improvement priorities	Gap Analysis Report	Capability Assessment and Risk Integration
6	IT Strategy Formulation	IS Strategy, IT Strategy, IS/IT Management Strategy	Architecture Vision and Requirement Management	APO02, APO05, EDM02: strategic alignment and value delivery	Develop IT strategy aligned with organizational strategy	IT Strategy Document	Strategic Alignment
7	Enterprise Architecture Design	Identify application and strategic information requirements	Phase B: Business Architecture; Phase C: Data and Application	APO03, APO14, BAI03	Transform IT requirements into target architecture blueprint	Enterprise Architecture Blueprint	Enterprise Architecture Blueprint

Commented [Bambang8]: Comment 5, Reviewer 2 :

1. The novelty claim is based primarily on combining three established frameworks rather than demonstrating a new integration mechanism. The manuscript provides extensive correspondence tables, but most mappings appear to place existing activities in sequence. It does not define conflict-resolution rules, feedback loops, formal traceability relations, optimization procedures, or theoretically derived constructs that could not be obtained by applying the source frameworks consecutively. The authors should identify the precise design knowledge contributed by PRASASTI, including how outputs are transformed between stages, how overlapping controls are reconciled, how priorities propagate through the architecture and governance layers, and which decisions differ from ordinary multi-framework tailoring.

Commented [Bambang8R2]: Thank you for this valuable and constructive comment. We agree that the previous manuscript did not sufficiently emphasize the specific design knowledge contributed by PRASASTI, which could make the proposed framework appear as a sequential combination of Ward and Peppard, TOGAF ADM, and COBIT 2019. To address this concern, we have revised the manuscript by clarifying that the novelty of PRASASTI is not based on merely integrating three established frameworks, but on introducing an integration mechanism that regulates the transformation, alignment, and traceability of outputs across strategic planning, enterprise architecture, and IT governance layers. (Table 6. Table 6. Integration Mapping of Ward and Peppard, TOGAF, and COBIT 2019 into the Proposed PRASASTI Framework)

The revised manuscript introduces an Integration Matrix Model of IT Strategic Planning Using Ward and Peppard, TOGAF ADM, and COBIT 2019. This model explicitly defines the contribution of each framework, the transformation mechanism between stages, the resulting artifacts, and the corresponding PRASASTI dimensions. In the proposed mechanism, Ward and Peppard is positioned as the strategic analysis foundation to transform business objectives into strategic IT requirements. These requirements are subsequently transformed through TOGAF ADM into enterprise architecture artifacts covering business, data, application, and technology domains. Furthermore, COBIT 2019 is integrated to translate architecture and strategic requirements into governance objectives, management objectives, control mechanisms, capability targets, and performance indicators. Unlike conventional multi-framework tailoring approaches that sequentially apply independent frameworks, PRASASTI introduces explicit transformation relationships: (1) requirement transformation from business objectives into IT strategic requirements; (2) priority propagation from business priorities into architecture and governance decisions; (3) artifact traceability from strategic requirements to architecture components, governance controls, and performance indicators; and (4) feedback mechanisms using capability assessment ... [2]

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No	IT Strategic Planning Stage	Ward & Peppard Contribution	TOGAF ADM Contribution	COBIT 2019 Contribution	Integration Mechanism / Transformation	Output Artifact	Dimension
			Architecture; Phase D: Technology Architecture				
8	IT Portfolio and Priority Planning	McFarlan Strategic application portfolio, investment priority	Phase E: Opportunities and Solutions	APO05, BAI01, BAI02	Determine application and program priorities based on business value, risk, and architecture requirements	IT Portfolio and Investment Priority	IS/IT Portfolio Planning
9	IT Governance Development	IS/IT Management Strategy	Architecture Governance	EDM01, EDM03, EDM04, APO01, APO13	Integrate governance structure, risk management, controls, and decision rights	IT Governance Framework	Governance Objective Mapping
10	Implementation Roadmap Development	Strategic initiatives and implementation planning	Phase E: Opportunities and Solutions; Phase F: Migration Planning; Phase G: Implementation Governance	BAI01, BAI11, APO05	Develop implementation stages based on priority, dependency, and organizational readiness	IT Implementation Roadmap	Performance Measurement and Implementation Planning
11	Investment and KPI Development	Strategic benefits and investment planning	Architecture Capability Management	MEA01, MEA04, APO06	Link IT investment with business benefits and performance indicators	IT Investment Plan and KPI Framework	Performance Measurement
12	Validation and Finalization	Strategic review and alignment evaluation	Architecture Governance Review	MEA01, MEA03: monitoring, evaluation, and compliance	Validate all artifacts to ensure alignment between strategy, architecture, and governance	Final IT Masterplan Document	Governance and Continuous Improvement

The table above shows that each framework occupies a complementary position. Ward and Peppard are used as the basis for strategic IS/IT analysis, primarily to identify business needs, IS/IT conditions, IS/IT strategies, and application portfolios. TOGAF is used to translate the results of this strategic analysis into enterprise architecture designs through the ADM phases, particularly Architecture Vision, Business Architecture, Data

Architecture, and Application Architecture. Architecture, Technology Architecture, Opportunities and Solutions, Migration Planning and Implementation Governance. Meanwhile, COBIT 2019 is used to strengthen governance, capability, risk, control, and performance measurement aspects through EDM, APO, BAI, DSS, and MEA processes.

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With this mapping, the new framework not only produces an IS/IT strategy but also provides an architectural blueprint, an implementation roadmap, and governance, objectives, capability assessment, risk control, KPI, and monitoring mechanisms. Methodologically, Ward and Peppard serve as the strategic foundation, TOGAF as the architectural foundation, and COBIT 2019 as the governance and measurement foundation

6.4. Demonstration

The demonstration stage was carried out using an expert-based framework walkthrough approach, employing an expert judgment mechanism and the Delphi method. At this stage, the developed framework artifact was demonstrated to the experts using a help document, the framework design, a flowchart, a table mapping components, and scenarios for using the framework. Experts were requested to examine how the framework can be used in the planning process of IT strategy, starting from strategic analysis.

Instrument questionnaire used to get an evaluation expert to clarify the concept, completeness of components, consistency of flow, relevance of indicators, and integration with inter-frameworks, as well as potential implementation of the framework in the organization. Assessment conducted across a number of Delphi rounds to reach consensus on the eligibility artifact. Input from experts in every field is used to refine the framework before entering the evaluation stage.

6.5 Evaluation

Stage fifth is Evaluation, namely evaluate artifacts that have been developed and demonstrated. In study this, evaluation can be done through expert judgment, Delphi method, validity test content,

6.6 Communication

The sixth stage is Communication, namely conveying the results of the study to community academics and practitioners through journal articles. In the journal article, stage communication is presented as a compilation of parts: introduction, literature review, research method, proposed framework, results and discussion, conclusions, limitations, and bibliography.

Results section: This section uses the results data collection based on a questionnaire to show the method framework and how to present the results.

Table 6. Profile respondents

Category respondents	Amount	Percentage
Planning expert IT strategy	5	26%

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Category respondents	Amount	Percentage
Enterprise architecture expert	5	26%
IT governance or COBIT expert	3	16%
IT Practitioner or IT manager	4	21%
Academics system information	2	11%
Total	19	100%

Table 6 above describes the profile of respondents involved in the planning framework assessment of the IT strategy. Total respondents: 19 people, across categories relevant to skills within an IT and governance organization. The respondent percentages presented in Table 6 are rounded to the nearest whole number for presentation purposes, while all statistical calculations were performed using the original numerical values.

Planning expert IT strategists and enterprise architecture experts each numbering 5 people (26%), emphasizing the role they play in evaluating alignment of strategy and enterprise architecture. There are 3 IT governance or COBIT experts (16%), providing perspective on IT governance, controls, and policies. IT practitioners or IT managers, numbering 4 people (21%), provide input on the practical, related implementation of the framework in real organizational environments. The academic system information totals 2 people (11%), adding a corner view of theoretical and scientific validity to the framework.

Next, testing related to validity and reliability, including content validity and reliability, and instrument planning framework assessment, IT strategic testing. This aims to ensure that each instrument item used is truly relevant, consistent, and reliable for assessing the framework scientifically. The results of the validity and reliability tests are seen in Table 7 below.

Table 7. Validity and reliability test results

Test	Results	Interpretation
CVR	0.68-1.00	Valid
S-CVI/Ave	0.98	Validity very good content
Aiken's V	0.79-0.94	Agreement expert tall
Cronbach's Alpha	0.89	Reliable
Kendall's W	0.74	Consensus strong
IQR Delphi round 2	<= 1	Consensus achieved

Commented [Bambang9]: This is for the added for comment 9 : (Reviewer 1)

Commented [Bambang10]: Comment 9 : (Reviewer 1)

The respondent percentages are rounded in a way that should be explicitly documented. The exact proportions are 26.3% for 5 of 19 respondents, 15.8% for 3 of 19, 21.1% for 4 of 19, and 10.5% for 2 of 19. Table 6 reports these as 26%, 16%, 21%, and 11%. The rounded values sum to 100%, so the table is not mathematically incorrect, but the manuscript should state that percentages were rounded to the nearest whole number and use the same precision consistently.

Commented [Bambang10R2]: Thank you for the comment. We have added an explanation that respondent percentages are rounded for presentation purposes, while all statistical calculations were conducted using the original numerical values. Has been added for the sentenced. below Respondent Profile Table, with :

The respondent percentages presented in Table 6 are rounded to the nearest whole number for presentation purposes, while all statistical calculations were performed using the original numerical values

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Based on Table 7, the instrument's validity and reliability testing indicate high reliability. CVR values range from 0.68 to 1.00, indicating that most items were deemed valid by the experts. S-CVI/Ave of 0.98 indicates that the validity of the content is a very good instrument. Aiken's V results show high agreement among experts (0.79–0.94), while Cronbach's Alpha of 0.89 indicates adequate reliability. Kendall's W value of 0.74 and the Delphi round 2 IQR ≤ 1 indicate strong consensus among experts on the suitability and eligibility of each instrument item.

Next, testing analysis against the readiness index and gap analysis for every planning framework dimension integrated with the IT strategy analysis. This aims to evaluate the organization's readiness to implement the framework and to identify the gap between the current condition and the ideal target for each dimension. As for the results, the readiness index testing and gap analysis are shown in Table 8 below. Testing analysis was conducted using the expert-assessed applicability index and gap analysis for each integrated IT strategic planning framework dimension. This evaluation aims to measure the perceived applicability and implementation feasibility of the PRASASTI framework artifact and identify the difference between the current expert assessment score and the defined evaluation target for each dimension.

Table 8. Readiness index and gap analysis. Expert-Assessed Applicability Index and Gap Analysis

Dimensions	Current/Target	Gap/Level
Strategic alignment	78% / 85%	7% / Managed
IS/IT portfolio planning	75% / 85%	10% / Managed
Enterprise architecture blueprint	72% / 85%	13% / Managed
Governance objective mapping	69% / 85%	16% / Managed
Risk and control integration	66% / 85%	19% / Managed
Capability assessment	70% / 85%	15% / Managed
Performance measurement	68% / 85%	17% / Managed
Implementation roadmap	73% / 85%	12% / Managed
Average	71% / 85%	14% / Managed

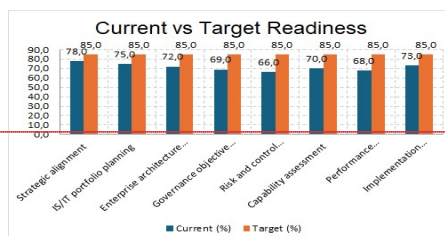


Figure 5. Current vs target readiness

Based on Table 8 and Figure 5, it can be seen that organizational readiness for all the framework dimensions is in the Managed category, the expert-assessed applicability level of all PRASASTI framework dimensions is categorized as Managed although there is a gap between dimension one and the others. The target value of 85% was defined as a benchmark for evaluating expert-assessed applicability of the framework. This threshold is used as an evaluation reference and should not be interpreted as a universal maturity standard applicable to all organizations. The Strategic Alignment dimension has a gap of 7% from current score of 78% and the target of 85%, while Risk and Control Integration has the largest gap, by 19% from current score of 66% and the target of 85%. Dimension: Other aspects, such as Enterprise architecture blueprint, Governance objective mapping, and Capability assessment, have a gap of around 13%, 16%, and 15%, while Performance measurement has a gap of 17%. Overall, the average readiness index across all dimensions reached 71.4% of the 85% target, with an average gap of 14%–15.6%, indicating that the organization is in a Ready Managed position but still needs several repairs to reach optimal conditions.

7. Discussion

Calculation results show that the planning framework integrated the IT strategy's own potential to strengthen the planning IT strategy by linking strategy, architecture, governance, risk, and performance into a single plot. Ward & Peppard provide a base analysis of business and IS/IT. TOGAF translates strategy into an enterprise architecture blueprint, while COBIT 2019 ensures that the design includes governance objectives, capability measurement, risk control, and KPIs.

The biggest gaps in the calculation are in risk and control integration and performance measurement. This shows that organizations often have planned applications and infrastructure but do not yet have

Commented [Bambang11]: Comment 10 : (Reviewer 1)

The CVR range of 0.68–1.00 is compatible with 19 experts only after rounding. Under the stated CVR formula, a value based on 16 of 19 experts rating an item as essential is approximately 0.684, while unanimous agreement produces 1.000. The authors should rep... [4]

Commented [Bambang11R2]: Thank you for this valuable comment. We have revised the presentation of all statistical indicators by applying a consistent rounding rule. The calculations were performed using the ori... [3]

Commented [Bambang11R3]: I have updated for the reason related this, at chapter 5. Research Methodology with, with sentences like this : ... [5]

Commented [Bambang14]: Comment 3 : (Reviewer 1)

3. The presentation of figures is not professional. In figures, letters are small and blurry. The authors sh... [8]

Commented [Bambang14R2]: Thank you for the constructive suggestion. We have revised all figures have been redesigned to improve readability and visual ... [9]

Commented [Bambang12]: Comment 2 (From Reviewer 2) : ... [6]

Commented [Bambang12R2]: For Our respond :

Thank you for this valuable clarification. We agree that the previous use of the term "readiness index" could ... [7]

Commented [Bambang15]: Comment 6 : (Reviewer 2)

The chosen 85% target and the "Managed" classifications are insufficiently justified and materially affect the ... [10]

Commented [Bambang15R2]: Thank you for the valuable comment. We agree that the previous manuscript did not sufficiently explain the rationale behind the target value and maturity classification ... [11]

Commented [Bambang13]: Comment 2 : (Reviewer 2)

Will be revised with "Table 8. Expert-Assessed Applicability Index and Gap Analysis"

Commented [Bambang16]: Comment 4 : (Reviewer 1)

The overall readiness index is reported inconsistently. The abstract gives an average readiness index of 7 ... [12]

Commented [Bambang16R2]: Thank you for identifying this inconsistency. We have carefully reviewed all reported values and standardized the presentation of the applicability index. The value is ... [13]

Commented [Bambang17]: Comment 7 : (Reviewer 1)

The manuscript should distinguish percentage-point differences from relative percentage gaps. For exa... [14]

Commented [Bambang17R2]: Thank you for this important clarification. Has been revised with red font color

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control over risks and indicators of strong performance. COBIT 2019 integration is important so that the IT roadmap is not only technical but also measurable, controllable, and sustainably evaluable.

In a theoretical sense, this research expands the draft planning of IT strategy by adding dimensions of enterprise architecture and IT governance. In general, this framework can be used by organizations to develop an IT Master Plan, a digital transformation roadmap, an enterprise architecture blueprint, and an IT governance improvement program.

7.1 Research Limitations and Future Research Directions

Although PRASASTI has been validated through expert judgment and Delphi evaluation, this study has several limitations. First, the evaluation was conducted at the framework artifact level and does not represent empirical evidence from organizational implementation. Second, the effectiveness of PRASASTI in improving IT strategic planning outcomes, decision-making processes, and organizational performance has not yet been measured through longitudinal studies. Third, the current validation focuses on content validity and expert consensus rather than operational performance evaluation.

8. Conclusion

This study proposes a new integrated model for planning an IT strategic framework, based on Ward & Peppard, TOGAF, and COBIT 2019. This framework was developed using the Design Science Research approach and was designed to be validated through expert judgment and the Delphi method.

Contribution to the main study: This is a development artifact conceptually linking analysis of strategic IS/IT portfolio application, enterprise architecture blueprint, governance objectives, capability assessment, risk control, implementation roadmap, readiness index, gap analysis, and performance measurement. This framework can serve as the basis for a more comprehensive IT Master Plan that is measurable and aligned with the organization's needs.

The study further recommended applying this framework to real-world organizations, such as colleges, high schools, hospitals, cooperative agencies, government agencies, and private-sector companies. Testing is empirically required to evaluate the framework's effectiveness in improving the quality of planning for IT strategy and IT governance.

Conflicts of interest

The authors declare no conflict of interest.

Author contributions

Conceptualization, BS, TS, and IR; methodology, BS; validation, TS, and IR; formal analysis, BS; investigation, B.S.; resources, BS; data curation, BS; writing-original draft preparation, BS; writing-review and editing, TS and IR; visualization, B.S.; supervision, TS and IR.

Acknowledgments

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Commented [Bambang18]: This is, updated for answer, comment 8, from Reviewer 2.

Commented [Bambang19]: Comment 8 : (Reviewer 2)

The conclusions overstate what the reported evidence establishes. The manuscript claims strengthened traceability, integrated risk control, continuous monitoring, and practical support for IT master planning, yet these properties are inferred from framework structure rather than measured through implementation. No traceability-completeness test, task-performance study, longitudinal monitoring result, or organizational outcome is presented. The conclusions should be restricted to preliminary conceptual development and expert-based content validation. Stronger claims require a real case implementation demonstrating end-to-end artifacts, decision changes attributable to PRASASTI, stakeholder outcomes, resource requirements, and post-implementation performance.

Commented [Bambang19R2]: Thank you for this valuable and constructive comment. We agree that the previous version contained several statements that could be interpreted as empirical evidence of organizational improvement, while the current study primarily focuses on artifact development and expert-based evaluation. Therefore, we have revised the conclusions and discussion sections by limiting the claims to conceptual development and expert validation outcomes. The revised manuscript clarifies that PRASASTI represents a validated design artifact that provides a structured mechanism for integrating IT strategic planning, enterprise architecture, and IT governance. Furthermore, we have added a research limitation section and future research roadmap emphasizing the need for real organizational implementation, longitudinal evaluation, and measurement of practical

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Page 1: [1] Commented [Bambang2R2] Bambang Suhartono 9/16/2026 3:19:00 PM

Thank you for the suggestion. We have revised the Abstract section by replacing approximate expressions with exact quantitative values obtained from the evaluation results.

Abstract: Information technology strategic planning is required to align digital investment , enterprise architecture , and governance technology information with the organization's objectives. However, the existing approach is still generally separated into strategic analysis, formulation architecture, and governance control. This research proposes an integrated framework based on Ward & Peppard, TOGAF, and COBIT 2019 using the Design Science Research approach. This research has never been conducted before. The framework is designed to connect analysis, business and systems information / technology information , portfolio applications , enterprise architecture blueprint , governance objectives, capability assessment, risk control, readiness index, and performance measurement. Validation was conducted using expert judgment and the Delphi method with 19 respondents. The calculation resultsshow that the framework is capable strengthen traceability between strategy, architecture , governance , risk , and implementation roadmap , where The strategic alignment dimension has a gap of 7% from ~~from current score of 78 % and a~~ the target of 85%, while risk and control integration has the largest gap . by 19% ~~from current score of 66 % and a~~ of the target of 85%, dimensions Other aspects , such as Enterprise architecture blueprint, Governance objective mapping , and Capability assessment have a gap of around 13%, 16% and 15%, while Performance measurement has a gap of 17%. ~~Overall, the average readiness index across all dimensions reached 71.4%, 79.4% of the target of 85%.~~ Overall, the average readiness index across all dimensions reached 71%of the target of 85%.

Contribution to the main study: These are conceptual artifacts that can serve as a basis for preparing the IT Master Plan and the organization's digital transformation.

Page 11: [2] Commented [Bambang8R2] Bambang Suhartono 9/16/2026 5:36:00 PM

Thank you for this valuable and constructive comment. We agree that the previous manuscript did not sufficiently emphasize the specific design knowledge contributed by PRASASTI, which could make the proposed framework appear as a sequential combination of Ward and Peppard, TOGAF ADM, and COBIT 2019.

To address this concern, we have revised the manuscript by clarifying that the novelty of PRASASTI is not based on merely integrating three established frameworks, but on introducing an integration mechanism that regulates the transformation, alignment, and traceability of outputs across strategic planning, enterprise architecture, and IT governance layers. (Table 6. Table 6. Integration Mapping of Ward and Peppard, TOGAF, and COBIT 2019 into the Proposed PRASASTI Framework)

The revised manuscript introduces an Integration Matrix Model of IT Strategic Planning Using Ward and Peppard, TOGAF ADM, and COBIT 2019. This model explicitly defines the contribution of each framework, the transformation mechanism between stages, the resulting artifacts, and the corresponding PRASASTI dimensions.

In the proposed mechanism, Ward and Peppard is positioned as the strategic analysis foundation to transform business objectives into strategic IT requirements. These requirements are subsequently transformed through TOGAF ADM into enterprise architecture artifacts covering business, data, application, and technology domains. Furthermore, COBIT 2019 is integrated to translate architecture and strategic requirements into governance objectives, management objectives, control mechanisms, capability targets, and performance indicators.

Unlike conventional multi-framework tailoring approaches that sequentially apply independent frameworks, PRASASTI introduces explicit transformation relationships:

- (1) requirement transformation from business objectives into IT strategic requirements;
- (2) priority propagation from business priorities into architecture and governance decisions;
- (3) artifact traceability from strategic requirements to architecture components, governance controls, and performance indicators; and
- (4) feedback mechanisms using capability assessment, gap analysis, and KPI evaluation for continuous improvement.

These revisions clarify that the contribution of PRASASTI is not the combination of existing frameworks, but the proposed integration mechanism that enables structured decision flow and traceability across IT strategic planning, enterprise architecture, and governance domains.

Page 14: [3] Commented [Bambang11R2] Bambang Suhartono 9/16/2026 5:46:00 PM

Thank you for this valuable comment. We have revised the presentation of all statistical indicators by applying a consistent rounding rule. The calculations were performed using the original values, while all reported results in tables and discussions are rounded to two decimal places for readability and consistency. This clarification has been added to the validation section.

Page 14: [4] Commented [Bambang11] Bambang Suhartono 9/16/2026 5:45:00 PM

Comment 10 : (Reviewer 1)

The CVR range of 0.68–1.00 is compatible with 19 experts only after rounding. Under the stated CVR formula, a value based on 16 of 19 experts rating an item as essential is approximately 0.684, while unanimous agreement produces 1.000. The authors should report CVR values to at least three decimal places or clearly state the rounding rule. The same precision policy should be applied to Aiken's V, Cronbach's alpha, Kendall's W, and readiness calculations.

Page 14: [5] Commented [Bambang11R3] Bambang Suhartono 9/16/2026 5:46:00 PM

I have updated for the reason related this , at chapter 5. Research Methodology with , with sentences like this :

All statistical indicators, including Content Validity Ratio (CVR), Content Validity Index (CVI), Aiken's V, Cronbach's Alpha, Kendall's W, and IQR, were calculated using the original numerical values. For presentation purposes, the results were rounded consistently to two decimal places to improve readability while maintaining calculation accuracy.

Page 14: [6] Commented [Bambang12] Bambang Suhartono 9/16/2026 5:38:00 PM

Comment 2 (From Reviewer 2) :

The reported readiness index is not demonstrably an evaluation of the proposed framework's effectiveness. The manuscript does not clearly identify whether experts assessed an actual organization, a hypothetical scenario, or the conceptual plausibility of the framework. Nevertheless, the results are interpreted as organizational readiness and assigned a "Managed" status. This creates a potentially fatal construct-validity problem because expert approval of framework dimensions is not equivalent to the readiness of an organization to implement them. The authors must define the unit of analysis, the assessed organization, the evidence used for every rating, the scoring rubric, and the meaning of each readiness category. Without an actual organizational evidence base, these results should be relabeled as perceived applicability or expert-assessed implementation feasibility.

Page 14: [7] Commented [Bambang12R2] Bambang Suhartono 9/16/2026 5:40:00 PM

For Our respond :

Thank you for this valuable clarification. We agree that the previous use of the term "readiness index" could be interpreted as measuring the actual readiness or maturity level of an organization, while the evaluation conducted in this study did not assess a specific organizational case. The previous terminology may therefore lead to a construct validity issue.

To address this concern, we have revised the terminology throughout the manuscript by replacing "readiness index" with "expert-assessed applicability index". The revised interpretation clarifies that the evaluation focuses on the PRASASTI framework artifact rather than organizational readiness. The unit of analysis in this study is the proposed framework, which was assessed by experts based on the completeness, relevance, consistency, feasibility, and applicability of its components.

Furthermore, we have revised the explanation of the scoring results and category interpretation. The obtained index value represents the experts' perception regarding the applicability and feasibility of implementing the proposed framework, and it should not be interpreted as evidence of organizational maturity, implementation readiness, or actual organizational capability.

The manuscript has also been revised in the Abstract, Methodology, Results, Discussion, and Conclusion sections to explicitly state this limitation and ensure that the conclusions remain aligned with **the Design Science Research** objective, which is **the development and evaluation of a conceptual** artifact.

We have Revised terminology throughout the manuscript.

Before : Table 8. Readiness index and gap analysis

After revised : Table 8. Expert-Assessed Applicability Index

Page 14: [8] Commented [Bambang14] Bambang Suhartono 9/16/2026 5:48:00 PM

Comment 3 : (Reviewer 1)

3. The presentation of figures is not professional. In figures, letters are small and blurry. The authors should enlarge or redraw figures. See Fig. 5. A part of X-axis label is hidden. The proposed presentation does not respect the elementary rules of a scientific writing

Page 14: [9] Commented [Bambang14R2] Bambang Suhartono 9/16/2026 5:48:00 PM

Thank you for the constructive suggestion. We have revised all figures have been redesigned to improve readability and visual quality. The revised figures use higher resolution formats,

Page 14: [10] Commented [Bambang15] Bambang Suhartono 8/11/2026 5:50:00 PM

Comment 6 : (Reviewer 2)

The chosen 85% target and the "Managed" classifications are insufficiently justified and materially affect the interpretation of all gap results. The paper applies the same target to heterogeneous dimensions such as strategic alignment, risk control, architecture, and performance measurement, without theoretical or empirical justification. It also assigns all observed scores to the same category despite substantial variation. The authors should define validated thresholds before data collection, justify whether dimension-specific targets are necessary, and conduct sensitivity analysis showing how conclusions change under alternative targets and weighting schemes. Percentage-point gaps should be distinguished consistently from relative percentage deficits.

Page 14: [11] Commented [Bambang15R2] Bambang Suhartono 8/11/2026 5:57:00 PM

Thank you for the valuable comment. We agree that the previous manuscript did not sufficiently explain the rationale behind the target value and maturity classification. In the revised manuscript, we have clarified that the 85% target value is not intended as a universal organizational maturity standard, but rather as an evaluation benchmark to assess the applicability and completeness of the proposed PRASASTI framework based on expert judgment.

Page 14: [12] Commented [Bambang16] Bambang Suhartono 9/16/2026 5:58:00 PM

Comment 4 : (Reviewer 1)

The overall readiness index is reported inconsistently. The abstract gives an average readiness index of 71.4%, Table 8 reports 71%, and the discussion again reports 71.4%. Based on the eight dimension scores in Table 8, the arithmetic mean is 71.375%, which may reasonably be reported as 71.4% or 71% depending on the rounding rule. The authors should adopt one rounding convention and apply it consistently throughout the abstract, results, discussion, and conclusion.

Page 14: [13] Commented [Bambang16R2] Bambang Suhartono 9/16/2026 5:58:00 PM

Thank you for identifying this inconsistency. We have carefully reviewed all reported values and standardized the presentation of the applicability index. The value is consistently reported as 71.4% using one decimal precision throughout the abstract, results, discussion, and conclusion sections.

For decimal in percentage, will be erased. For index 71.4% will be 71%, 15.6% will be changed 14 %

Page 14: [14] Commented [Bambang17]	Bambang Suhartono	9/16/2026 5:55:00 PM
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Comment 7 : (Reviewer 1)

The manuscript should distinguish percentage-point differences from relative percentage gaps. For example, the difference between a current score of 66% and a target of 85% is 19 percentage points. Expressed as a relative shortfall from the target, it is approximately 22.4%, calculated as $(85 - 66) / 85$. The manuscript repeatedly describes these values simply as percentages, which makes the quantitative interpretation ambiguous. One definition should be selected and used consistently